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In [1]: import pandas as pd          #Step 1: Import Libraries
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [3]: df = pd.read_csv('C:/Users/hp/desktop/product_recommendation_data.csv') #Step
print(df.head())
```

	UserID	Product	Rating
0	U1	Mobile	1
1	U1	Headphones	3
2	U1	Keyboard	2
3	U2	Mobile	2
4	U2	Mouse	2

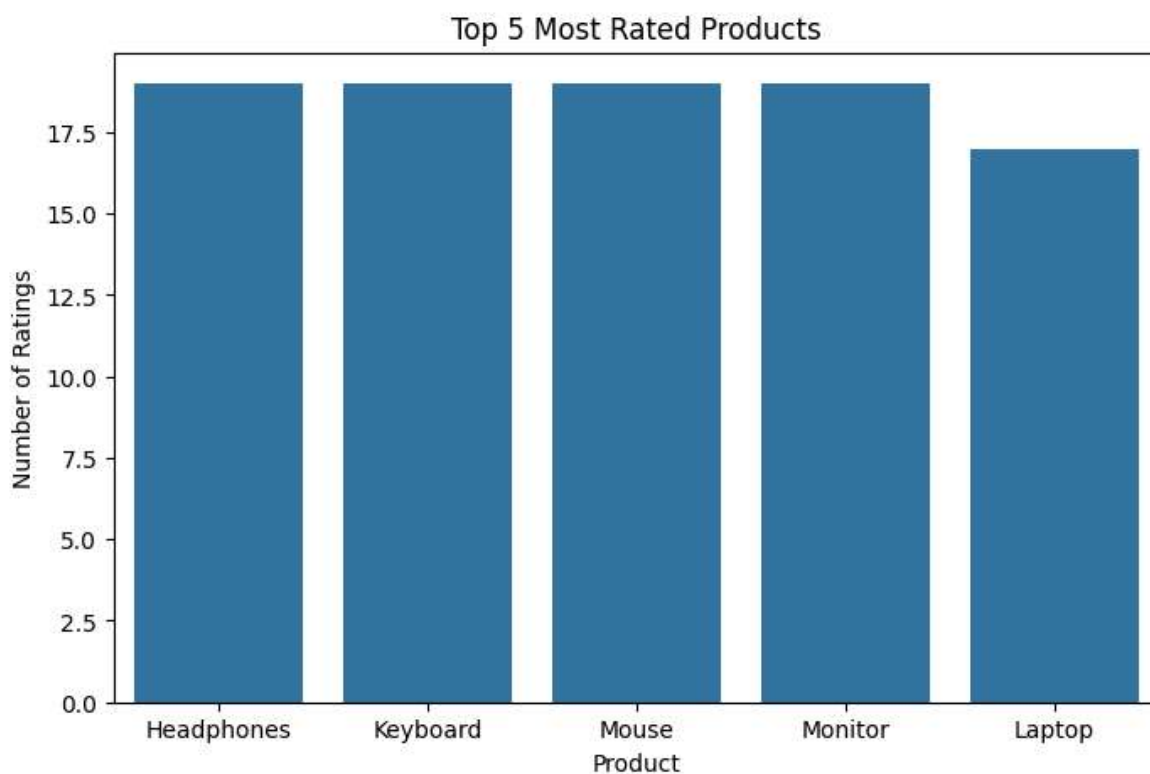
```
In [4]: print(df.info())          #Step 3: Basic EDA (Explore Data)
print(df['Product'].value_counts())
print(df['Rating'].value_counts())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 135 entries, 0 to 134
Data columns (total 3 columns):
#   Column   Non-Null Count  Dtype
---  -
0   UserID   135 non-null    object
1   Product  135 non-null    object
2   Rating   135 non-null    int64
dtypes: int64(1), object(2)
memory usage: 3.3+ KB
None
Product
Headphones    19
Keyboard      19
Mouse         19
Monitor       19
Laptop        17
Charger       17
Tablet        13
Mobile        12
Name: count, dtype: int64
Rating
3    35
5    34
1    29
4    20
2    17
Name: count, dtype: int64
```

```
In [5]: product_popularity = df.groupby('Product').agg({'Rating': ['count', 'mean']})
product_popularity.columns = ['Rating_Count', 'Average_Rating']
product_popularity = product_popularity.sort_values(by='Rating_Count', ascending
print(product_popularity)
```

	Rating_Count	Average_Rating
Product		
Headphones	19	3.157895
Keyboard	19	3.315789
Mouse	19	3.157895
Monitor	19	2.789474
Laptop	17	2.764706
Charger	17	3.352941
Tablet	13	3.076923
Mobile	12	3.166667

```
In [6]: top_products = product_popularity.head(5)           #Step 5: Plot Top 5 Products by
plt.figure(figsize=(8,5))
sns.barplot(x=top_products.index, y=top_products['Rating_Count'])
plt.title("Top 5 Most Rated Products")
plt.ylabel("Number of Ratings")
plt.xlabel("Product")
plt.show()
```



```
In [7]: def recommend_products(df, top_n=3):               # Step 6: Recommend Products (Simple
product_stats = df.groupby('Product').agg({'Rating': 'count'}).rename(column
top_products = product_stats.sort_values('Count', ascending=False).head(top_
return top_products.index.tolist()

# Recommend 3 products
recommended = recommend_products(df, top_n=3)
print("Recommended Products:", recommended)
```

Recommended Products: ['Headphones', 'Keyboard', 'Mouse']

In []: